

COMPSCI5079 (M)

Cryptography & Secure Development

Dr. Nguyen Truong

Lecturer (Assistant Professor), SoCS, University of Glasgow

Email: Nguyen.Truong@Glasgow.ac.uk

Office: SAWB 221C, Sir Alwyn Williams Building, 18 Lilybank Gardens



Lecture 1:

Fundamental Foundation for Cryptography





Lecture 1 - Review



Polynomial Arithmetic

- Exclusive OR is a useful operation in cryptography
 - Two applications of the same value cancel out.
- Polynomial arithmetic usually uses single bits (0 or 1) as coefficients of powers of x .
- This makes it easy to convert from bits to polynomials.
 - Each bit corresponds to a coefficient in the polynomial.
- It is common to split the bit string input into individual 8-bit bytes.
 - This involves arithmetic with polynomials with the highest power x^7 .
- Adding and subtracting polynomials is easy.
 - Each coefficient is combined using Exclusive OR.
 - If both coefficients are the same, the answer is 0.
 - If they are different the answer is 1.



Polynomial Arithmetic

- Multiplying polynomials produces longer polynomials.
- We can divide by a 'prime' (irreducible) polynomial and keep the remainder.
 - Like integers mod n .
- It is convenient to create a multiplication table so that multiplication calculations just need a fast table lookup.



Lecture Outline

1. Cryptography Overview & Terminology
2. Mathematical Foundation
 - Integer mod N
 - Polynomial Arithmetic
3. Information Theory



3. Information Theory



Information Theory

- Information theory examines situations where there are a lot of possible messages.
 - All messages are written in English.
 - All messages use the ASCII character set.
 - All possible images are made up of 1080 x 1920 pixels.
- Each message is made up of a series of letters.
- All possible letters form an alphabet.
 - All English letters
 - All English letters + punctuations.
 - All 64-bit values.



- $$H(X) = - \sum p(x_i) \log_2(1/p(x_i))$$



Entropy: Example

- Consider the `Gender` field in a database.
- The two possible values, `MALE` and `FEMALE` have equal probability.

$$p_1 = p_2 = 1/2$$

$$\begin{aligned} H &= 1/2 \log_2(2) + 1/2 \log_2(2) \\ &= 1/2 + 1/2 \\ &= 1 \end{aligned}$$

- Therefore 1 bit is needed to encode this information.
 - This is not a surprise!



Non-Equal Probabilities

- Now change the probabilities as follows:

$$p_1 = 7/8$$

$$p_2 = 1/8$$

$$\rightarrow H = 0.5410$$

- Fewer bits are needed when one value is more likely than the other.
 - We could group the values in 10's and use 6 bits per group.



Null Values

- Now assume that the database allows NULL values for the Gender field.
 - The gender information is not provided in half of the entries.

$$p_1 = p_2 = 1/4$$

$$p_3 = 1/2$$

$$\begin{aligned}\rightarrow H &= 1/4 \log_2(4) + 1/4 \log_2(4) + 1/2 \log_2(2) \\ &= 1/2 + 1/2 + 1/2 \\ &= 1.5\end{aligned}$$



Entropy: Another Example

- There are n messages, all equally likely.

$$p_i = 1/n \quad \text{for all values of } i.$$

$$\rightarrow H = -n \left(\frac{1}{n} \log_2(n) \right) = \log_2(n)$$

- $\log_2(n)$ is the number of bits in the binary representation of n .



The Redundancy of a Language

- All ways of encoding data have some structure which makes them different from purely random collections.
- English is highly structured, with a lot of **redundancy**.
- Compressing a file of English text reduces the structure, but there is still some structure in the compression encoding table.
- Information theory provides **quantitative estimates** of the **amount of redundancy**.
- Code-breaking attacks that exploit this redundancy are called **statistical attacks**.



The Absolute Rate of a Language

- The absolute rate (R) is defined as the **maximum number of bits** of information that could be encoded by the language, **assuming** all sequences of characters are **equally likely**.
- Let L be the number of characters in the alphabet.
- $R = \log_2(L)$ from the earlier example.
 - This means $L = 2^R$.
- In English, $R = \log_2(26) = 4.7$



The Actual Rate of a Language

- The actual rate of a language (r) is defined as the **average number of bits** of information **per character** in the language.
 - It is essentially the entropy per character.
- We can calculate it by defining the rate per character for all N character strings r_N and looking at the trend as N gets larger.
- We can define it in terms of entropy as:
$$r_N = H(X) / N$$
 - where X is the set of all messages of length N using characters in the language.



The Actual Rate of a Language

- As N increases, the rate decreases because there are fewer choices and some choices are more likely.
 - This decrease tapers off quickly to a constant value.
- In English, r_N is between 1.0 and 1.5 bits per character for large N .
- r is the value of r_N when N is large.
- This course will assume $r = 1.5$ for English.



The Redundancy of a Language

- The redundancy (D) of a language is defined as:

$$D = R - r$$

- In English, $D = 3.7 \dots 3.2$. We will use 3.2.
→ This corresponds to 68% redundancy.

- The redundancy comes from
 - Uneven single-letter distribution;
 - Uneven digram (2-letter) frequency;
 - Uneven trigram (3-letter) frequency.

- By deleting vowels and double letters:

mst ids cn b xprsd n fwr ltrs, bt th xprnc s mst
nplst.



Unicity Distance

- We are now in a position to ask an interesting question.
- If we decode some ciphertext to produce plaintext, how can we tell whether we have succeeded?
 - If the text reads as English rather than gibberish then we are part of the way there.
 - On the other hand, could we have produced a seemingly meaningful message by accident?
- How many characters of ciphertext do we need to be sure that a seemingly good decrypted message is the real one?
- This number is called the **unicity distance**.



Unicity Distance

- Assume we have an alphabet of $\mathbb{L}$ symbols.
- The number of **possible** messages of length N is:

$$\mathbb{L}^N = (2^R)^N = 2^{RN} \text{ since } \mathbb{L} = 2^R$$

- The number of **meaningful** messages can be expressed in terms of the rate for the language and is 2^{rN}
- If we assume that all messages are equally likely, then the probability of getting a meaningful message by chance is:

$$\begin{aligned} & \frac{\text{number of apparently meaningful messages}}{\text{number of possible messages}} \\ &= \frac{2^{rN}}{2^{RN}} = 2^{(r-R)N} = 2^{-DN} \end{aligned}$$



The Keyspace

- Try all possible keys and keep those that look right.
 - An **exhaustive search** or **brute force** attack.
- Let K be the set of all possible keys.
- Now the number of keys is $2^{H(K)}$
 - We must use the entropy of the keyspace, since the chosen key values may exhibit some redundancy.
- So, the expected number of wrong keys is:
 - $\circ = 2^{H(K)} - 1$, approximately $2^{H(K)}$
 - Where we assume there are a lot of possible keys, all but one of which is wrong.



The Keyspace

- The expected number of false solutions that look like they might be right:

(number of wrong keys) $\times$ (probability of chance meaningful message)

$$= 2^{H(K)} \times 2^{-DN} = 2^{H(K) - DN}$$

- This is a rapidly decreasing function of N .
- We can define the value of message length N at the crossover point when the exponent = 0 as the **unicity distance** N_u :

$$H(K) - DN_u = 0 \quad (\text{crossover point, exponent} = 0),$$

$$\rightarrow \mathbf{N_u = H(K) / D} \quad (\text{unicity distance})$$



Unicity Distance

- If $N > N_u$ then the chance of getting a false positive is negligible since the number of correct-looking false messages is much less than 1.
- On the other hand, if $N < N_u$ then many of the correct-looking messages will be false.
- N_u is roughly the number of characters needed to unambiguously break the code and determine the key.



Data Encryption Standard (DES) : Example

- The DES encryption algorithm has a key length of 56 bits.
If the keys are chosen at random then the entropy $H(K) = 56$.
- Assume an English language document is encrypted and that one byte is used for each letter.
 - Thus, $R = 8$ and hence $D = R - r = 6.5$.
 - Hence, $N_u = H(K) / D = 56 / 6.5 = 8.6$
- So, we need to decode about 9 letters to make sure that a message that looks like English is the real message.



Reducing Redundancy

- Reducing the value of D increased the unicity distance, and so makes it harder to break the code.
 - More characters are needed before an unambiguous solution is found.
- If the plain text were nearly random, and so D is very small, then the real message would be hard to tell apart from the actual random text.
- Compressing a message before encrypting will increase the unicity distance.



The Random Assumption

- The preceding model assumes a random encryption system.
- In particular, there is no relationship between:
 - similar keys and the ciphertext.
 - similar plaintext and the ciphertext.
- The unicity distance is less if the encryption is non-random.



Confusion and Diffusion

- This random concept is usually expressed in two terms: **confusion** and **diffusion**.
- An encryption system has good **confusion** if changing one bit in the **key** changes roughly half the bits of the ciphertext.
- An encryption system has good **diffusion** if changing one bit in the **plaintext** changes roughly half the bits of the ciphertext.



The usefulness of Unicity Distance

- The unicity distance does not provide a way of breaking the code.
- It just indicates how much information we need to gather before we can attempt to break it.
- It is based on the rate for a language, a statistical property, which is less valid for very short character strings.



Quizzes

1. Define the term "The entropy of a set of messages" and show how it can be calculated. A language contains 5 symbols: A, B, C, D and E. A, B, C each occur $\frac{1}{4}$ of the time, while D and E occur $\frac{1}{8}$ of the time. What is the entropy of this language?
2. Define the term "unicity distance." What information is needed to calculate it, and how useful is the concept of unicity distance? A newly invented language has 16 different symbols in its alphabet and is quite precise. On average each letter in the alphabet conveys 2 bits of information. A message in this language is encrypted with an 8-character key. It is known that users will choose English language keys all in lowercase. What is the unicity distance of these encrypted messages?



Lecture Summary



Information Theory

- The entropy of a system with many possible messages is the minimum number of bits needed to code all possible different messages.
 - It depends on the probability of each message occurring.

$$H(X) = \sum p(x_i) \log_2(1/p(x_i))$$

- This tells us how many items we need to search using a brute force attack (checking all possibilities)
 - $2^{H(X)}$



Information Theory

- If we know the number of bits used to convey any possible messages we can calculate
 - R , the absolute rate, assuming all messages are equally likely
 - r , the actual rate based on the probabilities of each message occurring in real life.
 - $D = R - r$, the redundancy of the message system
- If there is high redundancy then we can use a statistical attack, assuming a message is one of the likely ones.
- A brute force attack can produce possible plaintext messages which are wrong
 - They only look right by chance.
- The unicity distance tells us how many ciphertext letters we need to get to prevent these false positives.